Use-Case Flow Specification

Food Bank Surplus Redistribution Platform — Problem Statement #62

Use Case ID: UC-05
Use Case Name: Reserve Food Batch
Primary Actor: Shelter Coordinator
Supporting Actors: None (system automatically triggers UC-06 Generate Pickup Verification Code)

Preconditions

- The Shelter Coordinator is authenticated and verified (UC-01 completed).
- The selected food batch is active and has not expired.
- The selected food batch has available quantity remaining.

Postconditions

- The food batch's available quantity is decremented and, if fully claimed, marked "Reserved."
- The chosen pickup window is recorded against the reservation.
- A unique pickup verification code has been generated and shown to the Shelter Coordinator (via UC-06).
- The reservation and updated batch status are persisted in the system.

Main Success Scenario

1. The Shelter Coordinator opens the Browse Available Food screen.
2. The system displays all active, unexpired surplus food batches.
3. The Shelter Coordinator selects a suitable food batch.
4. The system displays batch details, including expiry countdown, quantity, dietary tags, and available pickup windows.
5. The Shelter Coordinator selects an available pickup window and confirms the reservation request.
6. The system re-validates that the batch is still active and has available quantity.
7. The system reserves the batch and decrements its available quantity.
8. The system triggers UC-06 to generate a unique pickup verification code.
9. The system records the reservation, pickup window, and verification code together.
10. The system displays a confirmation screen with the pickup window and verification code to the Shelter Coordinator.
11. The use case ends successfully.

Alternate Flow

A1. Batch becomes unavailable during reservation (triggered at step 6):
A1.1. The system detects that the selected batch has expired or its available quantity has reached zero since it was last displayed.
A1.2. The system rejects the reservation and does not decrement quantity or generate a code.
A1.3. The system informs the Shelter Coordinator that the batch is no longer available.
A1.4. The system redirects the Shelter Coordinator to the Browse Available Food screen with updated results.
A1.5. The use case ends without a reservation being created.